

DA-IICT

Template for Course Offering

1. Course Title	Printed and Flexible Electronics
2. Credit Structure	Lecture hours per week: 3 Tutorial hours per week: 0 Practical hours per week: 2 Total Credits: 4 (3-0-2-4)
3. Course Code	EL470
4. Program/Semester	UG (BTech 5 th Sem, BTech 7 th Sem) and PG (3 rd Sem)
5. Category	Elective
6. Prerequisite courses	Basic Electronic Circuits, Solid State Devices / Semiconductor Physics, Circuit Design Basics
7. Foundation for	This course lays the foundation for progressive work in flexible electronic devices and printed semiconductor technologies. It prepares students for careers in next-gen/emerging electronics, smart healthcare, and semiconductor industries.
8. Abstract Content	This course introduces ‘Flexible and Printed Electronics’ as a transformative extension of the semiconductor industry, enabling bendable, lightweight, and low-cost electronic systems. As the semiconductor industry evolves, flexible electronics plays a pivotal role in advancing wearables, IoT, and sustainable electronics. Students will explore key concepts in semiconductor devices, novel materials, and additive manufacturing techniques, gaining insights into the integration of flexible electronics for next-generation applications.
9. Course Materials/ References Suggested Text book(s)	1. M. M. Hussain and N. El-Atab, <i>Handbook of Flexible and Stretchable Electronics</i>, CRC Press, 2020. 2. M. Caironi, Y-Y. Noh, (Editors), <i>Large Area and Flexible Electronics</i>, 2nd Edition, WILEY-VCH, 2015. 3. A. M. Hussain, <i>Introduction to flexible electronics</i>. CRC Press, 2022. 4. W. S. Wong, and A. Salleo, (Editors) <i>Flexible electronics: Materials and Applications</i>, 1st Edition, Springer, 2009. Recent Research Papers on ‘Flexible and Printed Electronics’ area from scientific publisher like IEEE, IOP, ACS, AIP, Elsevier, RSC and others.
10. Evaluation/ Grading Policy	

Detailed Course Contents		
Topic Name	Content (2 -3 lines per 4 – 6 lectures)	No. of lectures
Introduction to Flexible and Printed Electronics	Need for flexible and printed electronics, Evolution of flexible electronics, Comparison with rigid electronics, Review of cutting edge research on flexible, conformable or printed electronics, Current trends and industrial/commercial applications (IoT, wearables, healthcare, agricultural etc.).	4
Materials for Flexible Electronics, Printing Techniques and Process Integration	Various semiconductors, dielectric and conducting materials, Examples of printable functional materials, Basics of charge injection and transport mechanism in flexible electronics, Thin-film deposition and processing methods for flexible devices, Solution-based Processing and Printing Technologies; Screen-printing technique, ink-jet printing, gravure, 3D printing and other processes.	12
Flexible Devices and Circuit (Thin-Film Transistors, Printed Circuit Boards and Sensors)	Thin-Film Transistors (TFTs); Device structure and performance, Electrical characteristics and parameters, Characterization methods for rigid and flexible devices, Electrical stability, Printed transistors, Transistors to Circuits; Flexible printed circuit boards (PCB), Flexible hybrid electronic systems, Fully flexible systems, Flexible Sensors; Temperature, gas/analyte, humidity, pressure and strain, Contacts and interfaces to flexible electronic devices.	14
Other Flexible Devices, System Integration and Reliability Testing	Organic Light Emitting Diodes, flexible OLED displays and lighting; Organic solar cells, Flexible Energy Storage Devices; Batteries and supercapacitors, RFID tags, Integration with wearables and IoT devices, Integration issues, Designs for the future and Reliability Testing; Mechanical, electrical and environmental reliability tests.	10
Outcomes and Objectives	Objective: This course aims to provide a solid foundation in flexible electronics, focusing on semiconductor devices, circuits, and materials. This course bridges traditional semiconductor fundamentals with next-generation electronics for wearables, IoT, and beyond. The course covers the fundamentals of flexible electronics, printing techniques for developing flexible electronic devices and circuits along with system integration for real-world applications. Outcome: This course aligns with the modern semiconductor industry and research trends. Through bridging the gap between traditional semiconductor fundamentals and modern/next-generation real-world applications, the course will help in preparing students to lead in both industry innovation and academic research in flexible and printed electronics.	
Comments		

Course Outcome:

CO1: Understand the principles of flexible electronics, including materials, processes and fabrication techniques for flexible electronic devices and circuits, with a focus on integration with traditional semiconductor systems.

CO2: To develop flexible electronic devices such as thin-film transistors, printed circuits, sensors and other flexible devices, applying knowledge of process integration and reliability evaluation.

CO3: Apply flexible electronics technologies to real-world applications in wearables, IoT, and beyond, while addressing challenges in performance, scalability, and system integration.

POs-COs Matrix:

POs-COs	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
CO1	✓				✓		✓					✓
CO2	✓	✓	✓		✓						✓	✓
CO3		✓		✓	✓	✓	✓			✓		✓

Flexible and Printed Electronics Lab

Lab Experiments

1. Analysis of surface properties in flexible substrates for printed electronics.

Objective: To evaluate surface wettability for printed ink adhesion on flexible substrate using contact angle measurement.

Outcome: Understand surface energy relevance in ink spreading and adhesion.

2. Development of conductive ink for flexible electronics and printing techniques.

Objective: To prepare a basic carbon or silver-based ink and test its properties.

Outcome: Learn principles of ink optimization for printing.

3. Printing resistive and/or capacitive tracks using conductive ink on flexible substrate to evaluate their electrical behavior.

Objective: To design and print resistive and/or capacitive structures on flexible substrates using conductive ink, and evaluate their electrical behavior.

Outcome: Understand material resistive/capacitive behavior along with tracking geometry effects.

4. Characterization of printed resistor using temperature-dependent resistance measurement to demonstrates TCR (temperature coefficient of resistance) using 2-wire method.

Objective: To measure resistance of printed track as a function of temperature.

Outcome: Demonstrates TCR (temperature coefficient of resistance) using 2-wire method.

5. Electro-physical analysis of moisture-induced conductance in printed tracks.

Objective: To investigate how moisture exposure affects the resistance of printed conductive tracks.

Outcome: Hands-on experience with resistive sensor characterization along with understanding of environmental effects on printed electronics.

6. Development of flexible electronics circuitry using conductive ink.

Objective: To fabricate a simple flexible electronics circuit using conductive ink on a flexible substrate, and then assemble and test a basic electronic circuit.

Outcome: Develop an understanding of how flexible electronics can be used in real-world applications.

7. Bending and reliability testing of printed conductive tracks.

Objective: To study how repeated mechanical bending affects the electrical resistance of printed conductive tracks on flexible substrates.

Outcome: Understanding mechanical reliability of flexible and printed electronics.

8. Fabrication and characterization of a flexible strain sensor using conductive ink.

Objective: To fabricate a flexible strain sensor using conductive ink (e.g., carbon, graphene etc.) on a flexible substrate and study its response under applied strain.

Outcome: Students will be able to measure and analyze the change in electrical properties with strain, demonstrating the sensor's sensitivity and flexibility.

Explanatory Notes
Enter a discipline (see master list for guidance) and a first digit followed by xx: BTech 6 th Sem, MTech 2 nd Sem, MSc DS 2 nd Sem.
Please indicate the program and desired semester in which it is to be offered. Comment on what additional courses in that semester it should not conflict with.
Appropriate categories: technical elective, open elective (BTech), elective (other programs)
Indicate the subsequent courses for which the course would be needed.
The Instructor is permitted a certain degree of flexibility in each course. The following ratios of primary to optional content in terms of lecture hours are given as guidelines: Foundation Courses – 75:25, Track Core (Group Elective)/M.Tech. Specialization Core – 66:33, Electives – 50:50. For example, a 3-0-0-3 course is expected to have 40-42 lectures in the semester, and the primary content should specify 30 lecture hours worth of material in a Foundation Level course and only 20 hours of material in an Elective Course. The primary content is what the instructor must cover in the course; the remaining is at the discretion of the Instructor.
The outcomes and objectives should give on the one hand the skills and competencies acquired by the student as well as the value of the course towards industry/research/academic objectives.